Supplementary material for
A spectral theory for turbulent burning velocity across scales

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This document accompanies the paper. Sections, figures, tables and equations numbered with the prefix S belong to this document; all other references are to the paper. The figures are produced by the notebook `make_figures.ipynb` from the same data as the figures of the paper.

S1 Distribution of the wrinkling exponent

The paper connects its burning-velocity relation with the power-law wrinkling factor of Charlette *et al.* (2002*a*) for an exponent $\beta_w = 1/2$. Charlette *et al.* (2002*b*) determined β_w locally with a dynamic procedure, and figure S1 shows the probability density function (PDF) of the exponents they obtained, digitised from their paper. The digitised curve, which ends at $\beta_w = 1$, was normalised to unit area and its moments were computed by trapezoidal integration. The mean is 0.587 and the standard deviation 0.104 (variance 0.0109); the median is 0.57 and the interquartile range 0.51–0.65. About 22 % of the distribution lies below $\beta_w = 0.5$, and the digitised PDF falls to zero at $\beta_w = 0.4$, above the fractal value $\beta_w = 1/3$ that would recover Damköhler's large-u' limit $s_T \propto u'$. The distribution therefore supports an exponent close to $1/2$, centred slightly above it.

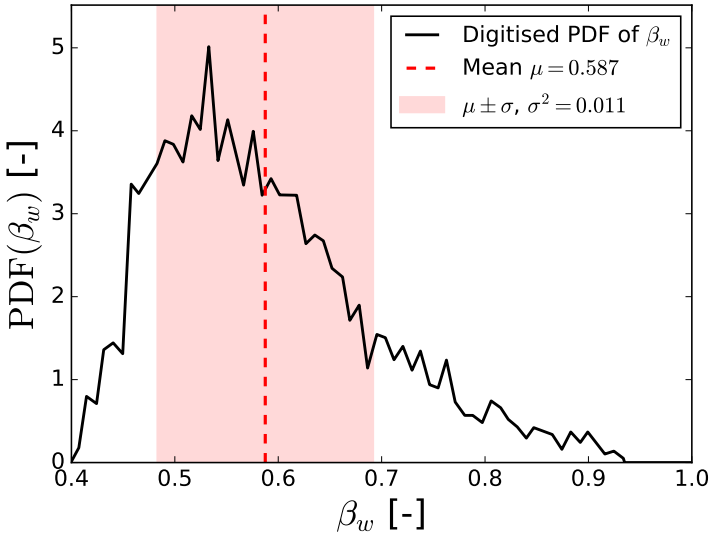


Figure S1: PDF of the local wrinkling exponent β_w obtained with the dynamic procedure of Charlette *et al.* (2002*b*) (digitised). Dashed line, mean $\mu = 0.587$; shading, $\mu \pm \sigma$, where $\sigma^2 = 0.011$ is the variance.

S2 Scaling of the burning velocity with Reynolds and Karlovitz numbers

Figure S2 shows the measured s_T/s_L divided by Ka against Re for the eight datasets of table 1 of the paper, with the same values and symbols. The dashed lines are $100\,Re^{-n}$, labelled with their slope $-n$.

Since $Ka = (u'/s_L)^{3/2}(\ell_0/\delta_L)^{-1/2}$ and $Re = (u'/s_L)(\ell_0/\delta_L)$, a series of measurements at fixed ℓ_0/δ_L in which $s_T/s_L \propto (u'/s_L)^m$ follows

$$\frac{s_T/s_L}{Ka} \propto Re^{\,m-3/2} \tag{S1}$$

in this plane. A slope of $-1/2$ thus corresponds to Damköhler's linear scaling, $m = 1$, and a slope of -1 to $m = 1/2$, the strong-turbulence limit of the paper's relation $s_T/s_L = [1 + C\,\mathcal{F}(Re)\,u'/s_L]^{1/2}$ apart from the weak variation of $\mathcal{F}$ with Re. Datasets in which ℓ_0/δ_L varies together with u'/s_L do not follow (S1) exactly.

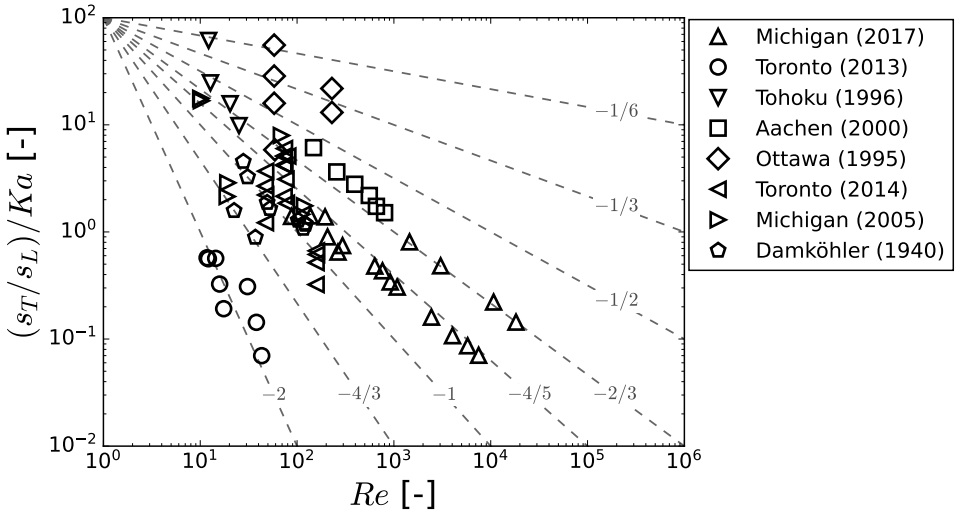


Figure S2: Measured s_T/s_L divided by Ka against Re for the datasets of table 1 of the paper (symbols as there). Dashed lines, $100\,Re^{-n}$, labelled with their slope $-n$.

S3 Fits to the spectral eddy viscosity and diffusivity

The functions $\nu_e^+(\kappa/\kappa_c)$ and $D_e^+(\kappa/\kappa_c)$ computed by Chollet (1984) were digitised for $0.01 \leqslant \kappa/\kappa_c \leqslant 1$ and fitted by nonlinear least squares with the form that Pope (2000) used for ν_e^+,

$$\nu_e^+,\ D_e^+ = C^{-3/2}\left[A + B\exp\left(-c\,\frac{\kappa_c}{\kappa}\right)\right], \qquad C = 1.5. \tag{S2}$$

Table S1 lists the coefficients and the values at the cusp, $\kappa = \kappa_c$, which are the values used in the paper; the turbulent Schmidt number is $Sc_T = \nu_e^+/D_e^+$. The fitted curves are shown in figure 2(b) of the paper. The regression of Pope (2000) for ν_e^+ gives a cusp value 26 % lower than the fit to the curve of Chollet (1984).

	A	B	c	Value at $\kappa = \kappa_c$
ν_e^+, Pope (2000)	0.441	15.2	3.03	0.640
ν_e^+, fit to Chollet (1984)	0.4927	162.7	4.996	0.868
D_e^+, fit to Chollet (1984)	0.8717	949.5	5.908	1.88
$Sc_T = \nu_e^+/D_e^+$				0.462

Table S1: Coefficients of (S2) and values at the cusp.

References

Charlette, F., Meneveau, C. & Veynante, D. 2002*a* A power-law flame wrinkling model for LES of premixed turbulent combustion Part I: non-dynamic formulation and initial tests. *Combust. Flame* **131** (1), 159–180.

Charlette, F., Meneveau, C. & Veynante, D. 2002*b* A power-law flame wrinkling model for LES of premixed turbulent combustion Part II: dynamic formulation. *Combust. Flame* **131** (1), 181–197.

Chollet, J.-P. 1984 Two-point closure used for a sub-grid scale model in large eddy simulations. In *4th Symposium on Turbulent Shear Flows* (ed. L. J. S. Bradbury, F. Durst, B. E. Launder, F. W. Schmidt & J. H. Whitelaw), pp. 62–72. Springer Berlin Heidelberg.

Pope, S. B. 2000 *Turbulent Flows*. Cambridge: Cambridge University Press.